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# ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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## ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead when processing long videos, a cost driven by the very large number of visual tokens their frame-level encoding produces. Prevailing efficiency methods curtail this overhead by pruning or merging tokens according to importance or similarity; yet, in doing so, they largely neglect a dimension of video content that is central to its meaning, namely change and its turning points, and they provide no unified account of spatio-temporal relationships among tokens. We resolve this gap by advancing a single organizing principle: similarity identifies redundancy, whereas difference localizes the key events that carry a video’s narrative. Building on this principle, we introduce ST-SimDiff, a training-free framework that first assembles the visual tokens into a spatio-temporal graph in order to model their spatial and temporal associations within one representation. A parallel dual-selection strategy then operates over this graph: (1) similarity-based selection applies community detection to retain a small set of representative tokens that compress static, redundant content; and (2) temporal-difference-based selection precisely localizes content-changing points and preserves the tokens that encode key dynamic shifts. Coordinating these two paths lets ST-SimDiff represent both static and dynamic content with a minimal token budget. Extensive experiments across multiple long-video benchmarks and base models demonstrate that our method consistently outperforms state-of-the-art approaches while substantially reducing computational cost. Our code will be released upon acceptance.

## 1 INTRODUCTION

The rapid maturation of Large Language Models (LLMs) has propelled a corresponding advance in Large Vision-Language Models (LVLMs), which now exhibit remarkable competence across image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For video, contemporary LVLMs sample the input into a sequence of frames and encode each frame into hundreds—or even thousands—of visual tokens that are subsequently consumed by the LLM (Bai et al., 2025; Li et al., 2024). Effective as this paradigm is, its token count grows explosively with video duration and resolution, imposing prohibitive computational and storage burdens that sharply constrain the deployment of LVLMs in demanding regimes such as long-video analysis and real-time interaction (Fu et al., 2024b).

To confront this bottleneck, a body of work has sought to improve LVLM efficiency, and these efforts fall into two broad families. The first is importance-based pruning. As FastV (Chen et al., 2024) observes, the attention distribution over visual information is markedly redundant across the layers of an LVLM; consequently, visual tokens that receive low attention scores in the deeper layers can be discarded to lighten the computational load. The second family is similarity-based merging and selection, which exploits the strong redundancy that visual tokens exhibit both across adjacent frames and within a single frame: FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision-encoder level.

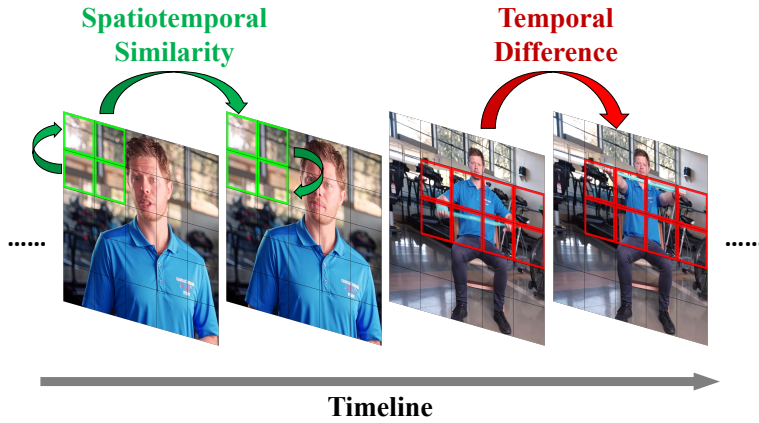


Figure 1: The core motivation of ST-SimDiff. We contend that efficient video understanding must address two complementary regimes at once: Spatiotemporal Similarity (left), which can be exploited to compress redundant information both spatially within a frame and temporally across adjacent frames; and Temporal Difference (right), which must be detected to capture the key actions and events that drive the plot.

Despite their success at visual token reduction, existing methods are limited in two decisive respects. First, they typically treat spatial correlations within a single frame and temporal correlations at fixed positions across frames in isolation; the absence of joint spatio-temporal modeling leaves them unable to characterize the complex dynamic events that structure real video. Second, and more fundamentally, they share a common blind spot: by optimizing almost exclusively for informational commonality—similarity and importance—they neglect the changes and differences that carry a video’s meaning. Because a video’s narrative is driven by turning events, a method that pursues similarity alone will smooth over a sudden action and, in turn, misread the content.

We resolve both limitations by adopting a single organizing principle: similarity identifies redundancy, whereas difference captures the key moments. An ideal video token compression algorithm, on this view, must satisfy two objectives at once: representing a video’s stable content with the fewest possible tokens, and precisely preserving its key changes. Guided by this principle, we propose **ST-SimDiff**, a dual token selection framework grounded in a spatio-temporal graph and composed of two complementary paths. The first is **similarity-based representative token selection**: treating every visual token as a node, we construct a graph over their feature similarity in the joint spatio-temporal dimension, apply a graph community detection algorithm to recover tightly connected token clusters—each corresponding to a stable, persistent visual element—and retain from each cluster the few tokens of highest centrality as its representatives. The second is **difference-based event token selection**: we attend specifically to the edges that run along the temporal axis, and wherever the similarity between corresponding tokens of adjacent frames drops sharply, we designate a turning point and mark the associated tokens as critical event tokens to be retained. The two token sets are finally merged and passed to the LLM. Our contributions can be summarized as follows:

- We are the first to establish that token similarity and token difference warrant equal standing in VLM efficiency research, recasting difference from an overlooked artifact into a first-class selection signal.
- We design a spatio-temporal graph that unifies the complex spatial and temporal relationships among video tokens within a single representation, and on it we build a dual token selection strategy that concurrently screens for both representative tokens and pivotal event tokens.
- Through extensive experiments across multiple long-video understanding benchmarks and base models, we show that ST-SimDiff sharply reduces the visual token budget while consistently maintaining—and in several settings improving upon—full-token performance.

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## 2 RELATED WORK

### 2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has substantially propelled multimodal understanding, giving rise to a family of prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These models encode images or video frames into visual tokens via pre-trained visual encoders and forward the tokens to an LLM, with an alignment module—commonly an MLP (Liu et al., 2024a;b) or a Q-Former (Li et al., 2023; Dai et al., 2023)—bridging the two. This architecture endows LVLMs with exceptional capability across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and it is especially promising for video understanding, where longer, higher-resolution, and more complex inputs must be processed (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This capability, however, comes at a pronounced computational cost, most acutely in video. Because continuous sequences of frames must be encoded, the number of visual tokens grows steeply and far exceeds that of static images, straining both training and inference efficiency. Efficiently processing this large volume of video tokens while reducing computational overhead therefore remains a central open problem in LVLM research.

### 2.2 VISUAL TOKEN COMPRESSION

Motivated by the substantial overhead of video processing, visual token compression has emerged as a central mechanism for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods approach compression along two principal axes: token importance and token similarity.

Earlier work concentrated on importance-based token pruning, which seeks to identify and discard the visual tokens that contribute least to model performance; such methods quantify importance through attention scores or other feature statistics, as exemplified by FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). A subsequent line of work (Shang et al., 2024; Ye et al., 2025) additionally incorporates token similarity to suppress redundancy more effectively while preserving critical information, recognizing that even high-importance tokens may be mutually similar and thus informationally redundant. FrameFusion (Fu et al., 2024b), for instance, combines similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information; and VisPruner (Zhang et al., 2025b) exploits visual cues to remove redundant items.

Despite this progress on visual redundancy, existing work scarcely addresses the intricate, global spatio-temporal similarity relationships among tokens. In video, visual information is redundant along both the spatial and the temporal axis simultaneously, yet similarity-driven methods such as FrameFusion attend chiefly to the temporal similarity between adjacent frames and leave the coupled spatio-temporal structure unexploited. This omission prevents them from fully harvesting the redundancy inherent in video and, in turn, curtails their performance under aggressive token reduction—precisely the regime our framework is designed to address.

## 3 PROBLEM DEFINITION

Given a video  $V$  and a text query  $Q$ , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of  $N$  visual tokens,  $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$ . This encoding is computationally demanding: the self-attention mechanism scales quadratically with the token count  $N$ , incurring  $O(N^2)$  complexity. Our objective is therefore to design an efficient token selection function  $f(\cdot)$  that maps the full sequence  $T_{\text{full}}$ , together with a compression ratio  $r$ , to a substantially smaller subset  $T_{\text{sub}} = f(T_{\text{full}}, r)$ , so as to maximize the LVLM’s downstream task performance under the budget constraint that  $|T_{\text{sub}}|$  is approximately  $r \cdot N$ . The central challenge is thus to construct  $f(\cdot)$  such that the retained subset  $T_{\text{sub}}$  concentrates the most critical information available in  $T_{\text{full}}$ .

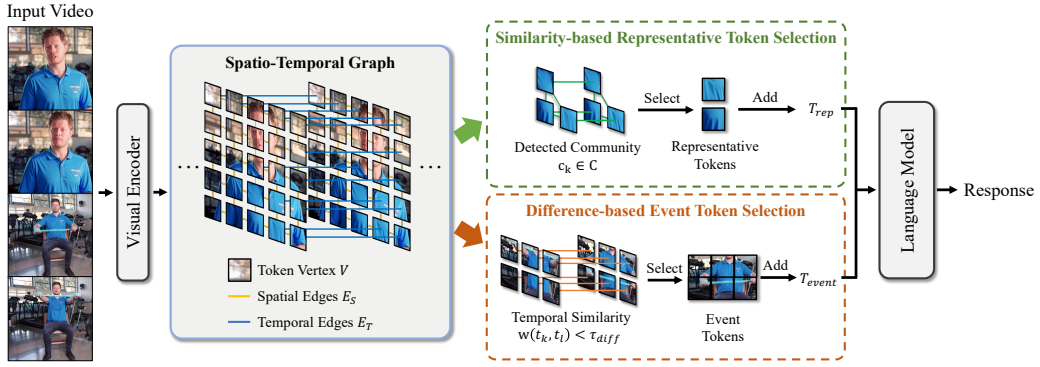


Figure 2: Overview of the ST-SimDiff framework, comprising three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

## 4 ST-SIMDIFF

### 4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff comprises two components that operate in parallel over a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The SRTS module identifies and condenses the video’s stable, redundant content: by applying a community detection algorithm to the graph, it groups highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and from each cluster it retains the few most central tokens to form the representative set  $T_{rep}$ . The DETS module, in complement, captures the video’s crucial dynamic shifts: it analyzes the temporal edges of the graph and locates the moments at which the similarity between corresponding tokens in adjacent frames drops sharply. Such points of high difference mark turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set  $T_{trans}$ . Combining the two paths, ST-SimDiff yields a final token subset  $T_{sub} = T_{rep} \cup T_{trans}$  that is at once highly compact and information-rich, retaining stable content while precisely capturing key events.

### 4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens  $T = \{t_1, t_2, \dots, t_N\}$ , where each token  $t_k \in T$  occupies a distinct spatio-temporal location and carries a feature vector  $\mathbf{x}_k$ . We specify this location through three indices: the frame index  $T(t_k)$ , the spatial height index  $H(t_k)$ , and the spatial width index  $W(t_k)$ .

We encode the relationships among these tokens in a spatio-temporal graph  $G = (V, E)$ , whose vertex set  $V$  comprises all tokens in  $T$  and whose edge set decomposes as  $E = E_S \cup E_T$ , isolating spatial and temporal relationships in two disjoint subsets. The spatial edges  $E_S$  connect tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges  $E_T$  connect tokens that occupy the same spatial position in adjacent frames, thereby encoding temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of every edge  $w(v_i, v_j) \in E$  is given by the cosine similarity between the feature vectors of its incident tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph compactly encodes both local spatial relationships and temporal continuity within a single structure, furnishing the unified foundation on which our dual-path token selection strategy operates.

#### 4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is inherently characterized by substantial spatio-temporal redundancy. In a static scene that persists for several seconds, for example, the background tokens remain highly similar over time; likewise, the tokens tracking an object that moves across the frame retain strong semantic correlation despite their shifting spatial positions. Whereas existing methods treat temporal and spatial similarity in isolation, our spatio-temporal graph is expressly constructed to capture their joint structure: within it, semantically related tokens coalesce into dense “communities” or “clusters” by feature similarity, irrespective of their particular frame or location. Selecting a representative from each community therefore constitutes a highly efficient strategy for compressing redundant information while preserving core semantics.

To identify these strongly correlated clusters precisely, we first sparsify the graph  $G$  by a similarity threshold  $\tau_{\text{sim}}$ , retaining only the edges whose weight exceeds it and thereby forming the pruned graph  $G' = (V, E')$  with  $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$ . We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to  $G'$  to recover the token clusters  $C = \{c_1, c_2, \dots, c_m\}$ . Within each community  $c_k \in C$ , we rank and filter tokens by centrality, where the centrality score  $S_c(t_a)$  of a token  $t_a \in c_k$  is its average similarity to all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We then set the intra-community retention rate to the globally specified compression ratio  $r$  and preserve, from each community  $c_k$ , the top  $\lceil |c_k| \cdot r \rceil$  tokens ranked by centrality. This uniform rule applies gracefully to every community, including single-node communities formed by unique tokens, for which the sole node is retained. Formally, the representative token set  $T_{\text{rep}}$  is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the  $k$  highest-scoring elements of a given set. In sum, by isolating and retaining the most central tokens within each semantic cluster, this strategy compresses the static and persistent content of the video efficiently and without discarding its representative structure.

#### 4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, difference defines its “events”. A model attuned solely to similarity may excel at recognizing “what is” yet falter on “what happened”, for a video’s plot is driven by key events whose essence is change—the appearance of a new object, the onset of an action, or a scene transition. In our formulation, such changes surface as a sharp drop in feature similarity between temporally adjacent tokens; accurately localizing these “turning points” of significant difference is therefore indispensable for understanding a video’s dynamic content and for correctly answering “when” and “why” questions.

Difference along the temporal dimension, in particular, reliably signals a key event: the entry of a new object, a scene change, or the beginning or end of an action induces a drastic change in the visual features at corresponding positions across adjacent frames. We accordingly analyze the temporal edges ( $E_T$ ) of the spatio-temporal graph. For a temporal edge  $(v_k, v_l) \in E_T$  joining two temporally adjacent tokens  $t_k$  and  $t_l$ , we adopt a dynamic threshold  $\tau$  (e.g., the 95th percentile of all temporal-edge difference scores); when the edge’s difference is large enough that  $w(t_k, t_l) < \tau_{\text{diff}}$ , we treat the subsequent token  $t_l$  as a critical event token and retain it. Formally, the event token set  $T_{\text{event}}$  is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This strategy preserves exactly those tokens that mark moments of significant temporal change; by capturing these key transitions, it guarantees that the dynamic aspects central to the video’s narrative are not lost during compression.

#### 4.5 OVERALL REDUCTION PROCESS

The framework first computes the representative set  $T_{\text{rep}}$  and the event set  $T_{\text{event}}$  in parallel and takes their union,  $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$ , as the initial candidate set. To meet the target token count  $N_{\text{target}} = \lceil r \cdot N \rceil$  exactly, we append a final pruning step: whenever  $|T_{\text{candidate}}|$  exceeds  $N_{\text{target}}$ , we discard the  $|T_{\text{candidate}}| - N_{\text{target}}$  least important tokens, where, following prior work (Chen et al., 2024), a token’s importance is read from its attention score in the shallow layers of the LLM. This step lets the method conform to any prescribed computational budget while retaining the most critical information.

In effect, ST-SimDiff integrates two complementary mechanisms: graph-based token selection and attention-based dynamic pruning. The former exploits the intrinsic structure of the video content—its similarity and its difference—to guarantee retention of the core information, while the latter provides a flexible control that meets any given budget precisely. This coupling renders the compression process at once principled and adaptive, achieving a robust balance between efficiency and performance.

#### 4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff decomposes across its three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Graph construction is the first stage. For each of the  $N$  visual tokens we compute the cosine similarity to its spatial and temporal neighbors; since every token has only a small, constant number of neighbors, this amounts to a single pass over all tokens and costs  $O(Nd)$ , where  $d$  is the token feature dimension.

SRTS then identifies the token clusters. As detailed in our implementation (Section 5.2), we employ a connected-components algorithm whose near-linear complexity is  $O(N+|E|)$ ; because  $|E| \leq 3N$ , this reduces to  $O(N)$ . The subsequent intra-community filtering could in principle be expensive—a naive all-pairs similarity computation within each community  $c_k$  would cost  $O(|C| \cdot |c_k|^2 \cdot d)$ —so we cap the maximum community size, partitioning any community that exceeds a predefined threshold (e.g.,  $|c_k| > \sqrt{N}$ ). This bound keeps the filtering step efficient and prevents it from dominating the overall cost. Although mainstream graph clustering algorithms such as the Louvain and Leiden methods admit richer community definitions, their higher complexity approaches  $O(N \log N)$ ; we therefore adopt the simpler, highly efficient connected-components algorithm to minimize processing overhead.

DETS performs a single pass over all temporal edges, comparing each edge’s token similarity against the threshold to detect significant changes, at a linear cost of  $O(Nd)$ .

Taken together, these linear-time operations govern the framework’s complexity, yielding an overall cost of  $O(Nd)$ . This is markedly more efficient than the quadratic  $O(N^2d)$  complexity of the LLM’s self-attention mechanism, so ST-SimDiff delivers substantial token reduction while adding only a negligible computational footprint relative to the model’s primary inference cost.

### 5 EXPERIMENTS

#### 5.1 EXPERIMENTAL SETUP

**Baselines** To evaluate our method comprehensively, we adopt two distinct base models, LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), and benchmark our approach against a representative spectrum of state-of-the-art video token compression techniques. These baselines span the field’s two dominant strategies: **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as

| Method                         | VideoMME    |             |             |             | LongVideoBench |             |             | EgoSchema   |
|--------------------------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                | Short       | Medium      | Long        | Overall     | Relation       | Perception  | Overall     |             |
| Upper Bound (Full Performance) |             |             |             |             |                |             |             |             |
| LLaVA-Video                    | -           | -           | -           | 63.3        | -              | -           | 58.2        | 57.3        |
| Token Retain Ratio $r = 30\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 68.2        | 58.6        | 51.2        | 59.3        | 48.0           | 59.9        | 53.5        | 51.3        |
| PruMerge                       | 70.2        | 57.3        | 52.2        | 59.9        | 49.6           | 60.5        | 54.7        | 50.9        |
| FasterVLM                      | 71.3        | 57.6        | 51.4        | 60.1        | 51.0           | 61.3        | 55.8        | 52.6        |
| VisionZip                      | 68.4        | 55.9        | 50.4        | 58.3        | 46.9           | 60.3        | 53.2        | 53.0        |
| FrameFusion                    | 74.0        | 59.8        | 50.0        | 61.3        | 49.7           | <b>63.3</b> | 56.0        | 53.0        |
| <b>ST-SimDiff (Ours)</b>       | <b>74.7</b> | <b>61.9</b> | <b>53.0</b> | <b>63.2</b> | <b>52.5</b>    | 63.2        | <b>57.5</b> | <b>56.0</b> |
| Token Retain Ratio $r = 50\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 73.9        | 61.4        | 51.3        | 62.2        | 50.1           | 62.2        | 55.7        | 54.7        |
| PruMerge                       | 72.3        | 58.7        | 52.8        | 61.3        | 51.4           | 63.2        | 56.9        | 54.6        |
| FasterVLM                      | 74.0        | 59.2        | 51.8        | 61.7        | 51.3           | 62.2        | 56.4        | 56.2        |
| VisionZip                      | 72.6        | 57.7        | 51.6        | 60.6        | 51.3           | 63.2        | 56.8        | 54.2        |
| FrameFusion                    | 74.6        | 61.2        | 52.0        | 62.6        | 51.7           | 64.2        | 57.6        | 55.8        |
| <b>ST-SimDiff (Ours)</b>       | <b>76.2</b> | <b>62.4</b> | <b>52.7</b> | <b>63.8</b> | <b>52.3</b>    | <b>64.3</b> | <b>57.9</b> | <b>57.3</b> |

Table 1: Performance comparison on long-form video understanding benchmarks with LLaVA-Video-7B at 64 input frames (%). For each metric, the best result among all compared methods is emphasized in **bold**. ST-SimDiff attains the top score in nearly every column at both the 30% and 50% token retain ratios.

VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For a fair and complete comparison, we further bracket these methods with the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equal number of tokens as a lower bound.

**Benchmarks** To probe our method’s capabilities in long-video understanding, we evaluate on three challenging benchmarks that jointly span foundational understanding, fine-grained retrieval and reasoning, and egocentric causal inference: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for foundational video understanding, encompassing diverse video types across three length categories: Short, Medium, and Long. LongVideoBench assesses the ability to retrieve and reason about details in long videos through a “referring reasoning” task, with videos at four durations: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering benchmark centered on long-term, first-person video that stresses the model’s grasp of character intentions and the causal chains linking actions.

## 5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs with 48 GB of VRAM under Ubuntu 22.04, and inference is evaluated with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video we retain its original setting of 64 input frames, and to keep computational efficiency and resource usage directly comparable, we fix NVILA’s input frame count to 64 as well. We set the similarity threshold to  $\tau_{\text{sim}} = 0.8$  and the difference threshold to  $\tau_{\text{diff}} = 0.2$ , and we report results at two token compression rates,  $r = 30\%$  and  $r = 50\%$ . In practice, weighing processing time against accuracy, we instantiate the community detection step with connected-component finding.

| Method                         | VideoMME    |             |             |             | LongVideoBench |             |             | EgoSchema   |
|--------------------------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                | Short       | Medium      | Long        | Overall     | Relation       | Perception  | Overall     |             |
| Upper Bound (Full Performance) |             |             |             |             |                |             |             |             |
| NVILA-Video                    | -           | -           | -           | 61.5        | -              | -           | 56.3        | 52.9        |
| Token Retain Ratio $r = 30\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 69.4        | 55.3        | 48.8        | 57.9        | 50.8           | 55.6        | 53.0        | 49.7        |
| PruMerge                       | 71.0        | 53.7        | <b>49.9</b> | 58.2        | 49.3           | 58.1        | 53.4        | 47.5        |
| FasterVLM                      | 72.6        | 56.9        | 51.0        | 60.1        | 48.7           | 57.8        | 53.0        | 49.3        |
| VisionZip                      | 71.4        | 56.2        | 49.7        | 59.1        | 46.3           | 56.2        | 50.9        | 48.9        |
| FrameFusion                    | 72.1        | 55.7        | 48.7        | 58.8        | 51.3           | <b>59.3</b> | 54.9        | 51.3        |
| <b>ST-SimDiff (Ours)</b>       | <b>73.3</b> | <b>57.7</b> | 49.7        | <b>60.2</b> | <b>51.7</b>    | 59.2        | <b>55.2</b> | <b>51.7</b> |
| Token Retain Ratio $r = 50\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 71.9        | 58.3        | 49.3        | 58.9        | 49.5           | 58.8        | 53.9        | 50.2        |
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| <b>ST-SimDiff (Ours)</b>       | <b>73.9</b> | <b>59.7</b> | <b>51.4</b> | <b>61.7</b> | <b>52.3</b>    | <b>61.3</b> | <b>56.5</b> | 52.5        |

Table 2: Performance comparison on long-form video understanding benchmarks with NVILA-Video-8B at 64 input frames (%). For each metric, the best result among all compared methods is emphasized in **bold**. The gains observed on LLaVA-Video-7B carry over to this distinct base model, evidencing the architecture-agnostic effectiveness of ST-SimDiff.

### 5.3 COMPARISON WITH STATE-OF-THE-ARTS

To evaluate ST-SimDiff comprehensively, we compare it against a range of mainstream efficient video language model (VLM) compression methods on three widely used long-form video understanding benchmarks—VideoMME, LongVideoBench, and EgoSchema. To establish the generalization and robustness of our method, we report detailed results on two distinct base models, LLaVA-Video-7B and NVILA-8B, at token retain ratios of 30% and 50%. The results appear in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms every competing method across all test configurations. Most tellingly, at the 50% token retain ratio its overall performance on both base models not only surpasses all other compression algorithms but, on several benchmarks, matches or exceeds the original model operating on 100% of the tokens—on LLaVA-Video-7B, for instance, ST-SimDiff reaches 63.8 on VideoMME against the 63.3 full-token upper bound. This result establishes the effectiveness and novelty of ST-SimDiff and motivates a closer analysis of the strategies and limitations of the baselines.

(2) Among the importance-based baselines (e.g., FastV, FasterVLM), FasterVLM is generally the stronger performer; their shared limitation, however, is an inefficient treatment of the temporal redundancy pervasive in video, since they tend to retain important yet repetitive tokens and thereby cap their compression efficiency. The hybrid baselines (e.g., PruMerge, VisionZip, FrameFusion) are more competitive than importance-only approaches, with FrameFusion typically the strongest overall. Even so, their common objective remains the identification and preservation of representative “commonality” information, leaving a systematic blind spot: the key narrative information carried by “turning points” and “abrupt events” is liable to be overlooked.

(3) In sum, the baselines reflect a progressively deeper grasp of “redundancy” in video, yet two core challenges persist: first, the absence of a synergistic analysis of the joint spatio-temporal similarity

| Method                        | VideoMME    | LongVideoBench | EgoSchema   |
|-------------------------------|-------------|----------------|-------------|
| Token Retain Ratio $r = 30\%$ |             |                |             |
| Baseline                      | 60.3        | 56.2           | 54.8        |
| + Sim                         | Spatial     | 61.5           | 56.5        |
|                               | Temporal    | 61.7           | 56.8        |
|                               | Spa. + Tem. | 62.6           | 57.0        |
| <b>++ Diff (Proposed)</b>     | <b>63.2</b> | <b>57.5</b>    | <b>56.0</b> |
| Token Retain Ratio $r = 50\%$ |             |                |             |
| Baseline                      | 63.2        | 56.7           | 56.5        |
| + Sim                         | Spatial     | 63.3           | 57.0        |
|                               | Temporal    | 63.7           | 57.2        |
|                               | Spa. + Tem. | 63.7           | 57.8        |
| <b>++ Diff (Proposed)</b>     | <b>63.8</b> | <b>57.9</b>    | <b>57.3</b> |

Table 3: Ablation of the framework’s components on LLaVA-Video-7B (%), reported at the 30% and 50% token retain ratios. Beginning from the importance-based Baseline, adding the similarity module (+ Sim)—and, within it, its spatio-temporal joint variant (Spa. + Tem.)—and then the difference module (++ Diff) each yields consistent gains, with the complete framework performing best across all benchmarks.

among visual tokens, which impedes the capture of complex redundancy; and second, a general neglect of “difference” detection, which fails to actively preserve the key changes that define plot development. ST-SimDiff’s performance gains follow directly from resolving both: it constructs a spatio-temporal graph that unifies these complex relationships and introduces difference detection as an explicit, first-class selection signal.

#### 5.4 ABLATION STUDY

To isolate the contribution of each core component of ST-SimDiff, we conduct a detailed ablation study. Starting from a Baseline that performs only fundamental importance-based pruning, we progressively add the similarity selection module (+Sim) and then the difference selection module (++Diff), quantifying each in turn. All experiments are run at token retain ratios of 30% and 50%, with results in Table 5.

(1) We first assess the +Sim module. Introducing similarity modeling on top of the Baseline consistently improves performance. Decomposing the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants reveals that jointly modeling the spatio-temporal associations among visual tokens is the most effective route to redundancy compression, confirming that spatial and temporal similarity must be treated together rather than in isolation.

(2) Adding the difference selection module (++Diff) on top of the optimal spatio-temporal joint similarity module completes the framework and produces a marked performance leap, most pronounced under the aggressive 30% compression ratio. This corroborates the module’s complementary role: whereas the aggressive +Sim module targets redundancy, the ++Diff module acts as a safeguard that identifies and preserves the unique event tokens defining the video’s narrative. Its marginal gain is smaller at the 50% ratio, where the more lenient +Sim module has typically already captured many of these event tokens and overall performance is nearing its upper bound. This pattern confirms our central motivation: the framework’s strength lies in balancing similarity-based redundancy compression against the essential preservation of difference-based events, a balance that matters most precisely in the demanding high-compression regime.

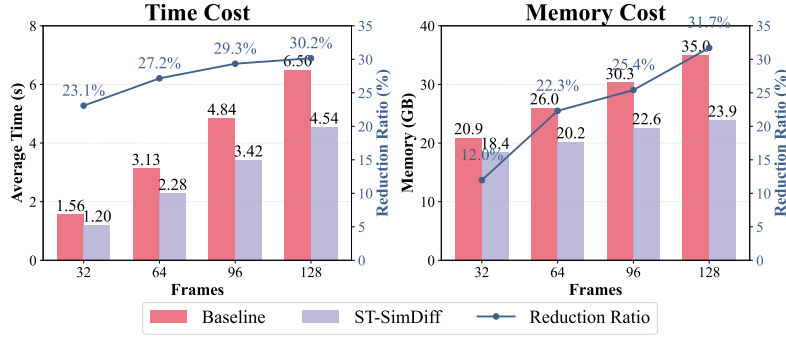


Figure 3: Computational cost of ST-SimDiff versus the uncompressed LLaVA-Video baseline, measured as average inference time and peak GPU memory usage under a 30% token budget across inputs of 32 to 128 frames. The efficiency advantage widens as the frame count grows.

## 5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency of ST-SimDiff, we measure its inference time and peak GPU memory usage under a 30% token budget and compare against the uncompressed baseline (Baseline LLaVA-Video). The results, reported in Figure 3 across input lengths from 32 to 128 frames, isolate the runtime and memory benefits of our compression at a fixed budget.

**Inference Time** ST-SimDiff delivers a substantial and scaling speedup. As the input grows, the baseline’s average inference time climbs steeply from 1.56 seconds (32 frames) to 6.50 seconds (128 frames), whereas our method curtails this growth to only 4.54 seconds at 128 frames. Crucially, the time saved grows with video length, rising from 23.0% at 32 frames to 30.2% at 128 frames—precisely the regime in which efficiency matters most.

**Memory Cost** ST-SimDiff yields comparably strong gains in memory. The baseline’s peak GPU memory grows linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames); by sharply reducing the number of tokens fed to the LLM, our method relieves this pressure, holding peak usage to 23.9 GB on 128-frame videos—a 31.7% saving over the baseline. This reduction enables the processing of longer videos on memory-constrained hardware, broadening the settings in which the model can be deployed.

Taken together, ST-SimDiff attains leading accuracy while simultaneously offering pronounced advantages in both inference time and memory, establishing it as an efficient solution for long-video understanding.

## 6 CONCLUSION

This paper confronts the computational inefficiency of Multimodal Large Language Models on long video—a problem rooted equally in the excessive number of visual tokens and in the systematic neglect of critical content change. We propose ST-SimDiff, a training-free framework that assembles the visual tokens into a spatio-temporal graph and models their complex associations within a single representation. On this graph a dual selection strategy operates: similarity-based selection identifies redundancy through community detection, while temporal-difference-based selection preserves the tokens at key transition points. Extensive experiments across multiple long-video benchmarks and two distinct base models show that ST-SimDiff substantially reduces both processing time and memory usage while consistently surpassing state-of-the-art performance—in several settings matching or exceeding full-token accuracy. More broadly, by elevating difference to equal standing with similarity, this work opens a new direction for efficient video representation, one that balances the compression of stable content against the preservation of the transitional events that carry a video’s meaning.

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## A VISUALIZATION OF THE TOKEN SELECTION PROCESS

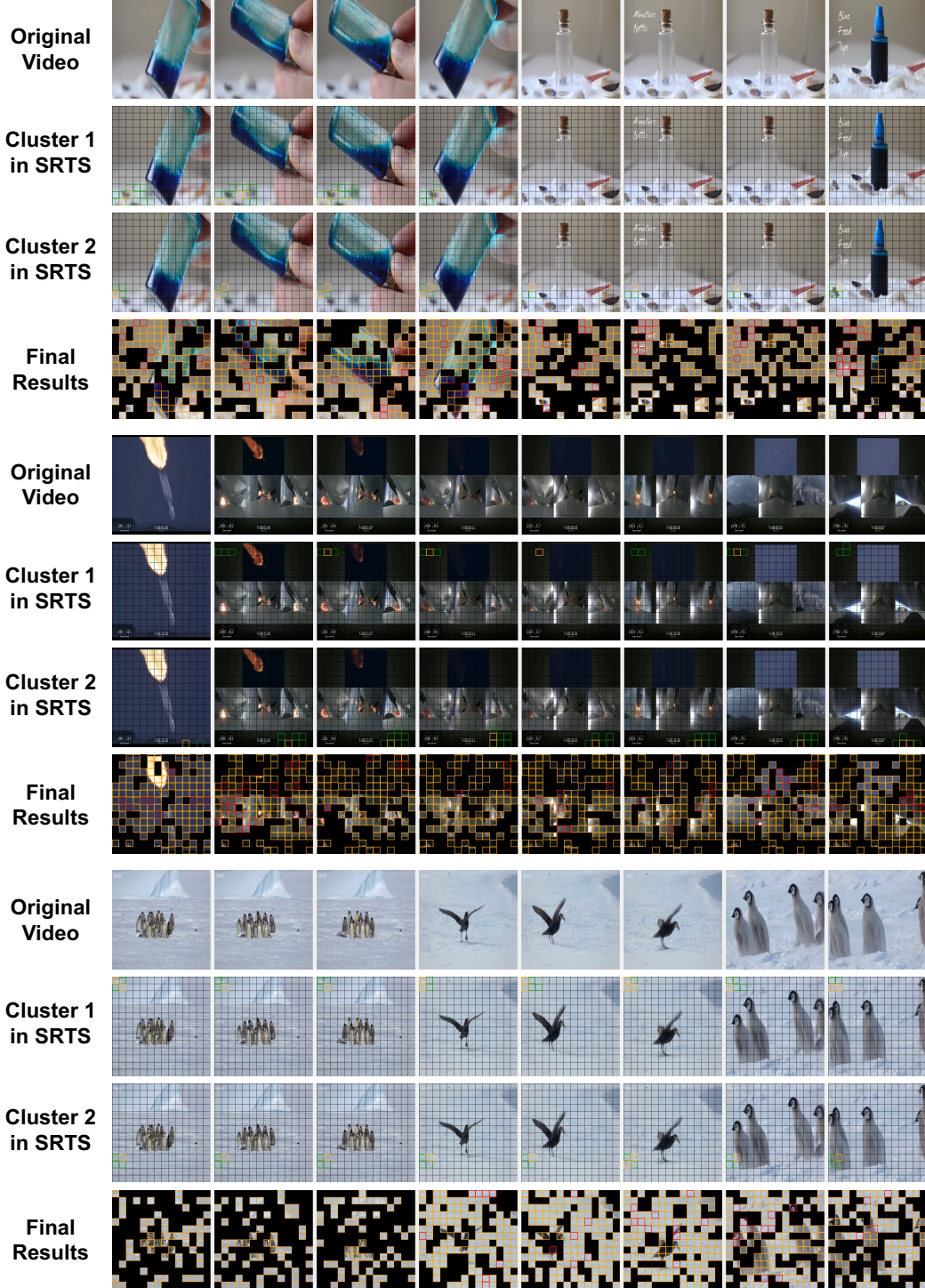


Figure 4: Visualization of the ST-SimDiff process. Token selection is decomposed into the communities detected by SRTS (shown as grids in Cluster 1 and Cluster 2) and the event-driven tokens detected by DETS. The final result makes the synergy explicit: sparse representative tokens (yellow) encode stable content, whereas dense event tokens (red) capture dynamic actions.

To render the operation of ST-SimDiff concrete, Figure 4 presents representative samples of the intermediate and final token selections on a video sequence depicting a dynamic object-manipulation task. Consider first the Similarity-based Representative Token Selection (SRTS): the rows labeled “Cluster 1” and “Cluster 2” show the graph community detection algorithm at work, grouping spatially and temporally redundant regions—such as the static background and stationary objects—into cohesive communities. The visualization confirms that SRTS reliably isolates stable visual elements and compresses them into a few central “representative tokens,” which appear as sparse yellow boxes in the final results, allowing the model to absorb massive background redundancy at minimal cost.

Complementing SRTS, the Difference-based Event Token Selection (DETS) is decisive at dynamic moments. In the “Final Results” row, the tokens triggered by high temporal difference are marked with dense red bounding boxes that coincide with the moving hand and the shifting blue liquid, confirming that our method precisely localizes the “turning points” at which the temporal difference crosses the threshold ( $\tau_{diff}$ ). This preserves fine-grained action detail and rapid visual change, so that the model captures the essence of the event rather than being swamped by static content.

The synergy of the two components is most evident in the overlay of the final results, where the balance is direct: sparse yellow tokens represent the stable background and dense red tokens capture the rapid motion. This contrast is visual evidence for our core motivation—ST-SimDiff avoids the blind spot of similarity-only approaches by sampling densely through rapid change, while at the same time pruning the spatiotemporal redundancy that importance-based methods routinely retain. The framework thereby strikes a favorable balance between computational efficiency and the preservation of critical visual information.

## B COMPARISON RESULTS ON QWEN2.5-VL

| Method                         | VideoMME |        |      |         | LongVideoBench |            |         | EgoSchema |
|--------------------------------|----------|--------|------|---------|----------------|------------|---------|-----------|
|                                | Short    | Medium | Long | Overall | Relation       | Perception | Overall |           |
| Upper Bound (Full Performance) |          |        |      |         |                |            |         |           |
| Qwen2.5-VL-64                  | 74.2     | 62.7   | 51.8 | 62.9    | 54.2           | 64.8       | 59.2    | 63.0      |
| Qwen2.5-VL-128                 | 77.4     | 65.8   | 56.1 | 66.4    | 56.2           | 65.1       | 60.4    | 63.9      |
| Token Retain Ratio $r = 30\%$  |          |        |      |         |                |            |         |           |
| ST-SimDiff-64                  | 73.8     | 61.7   | 51.8 | 62.4    | 52.2           | 62.1       | 56.8    | 62.7      |
| ST-SimDiff-128                 | 75.4     | 64.7   | 55.2 | 65.1    | 53.1           | 65.0       | 58.6    | 64.4      |
| Token Retain Ratio $r = 50\%$  |          |        |      |         |                |            |         |           |
| ST-SimDiff-64                  | 74.2     | 62.1   | 52.2 | 62.9    | 54.2           | 63.2       | 58.4    | 62.7      |
| ST-SimDiff-128                 | 76.8     | 66.3   | 55.8 | 66.3    | 54.9           | 65.3       | 59.8    | 64.5      |

Table 4: Performance comparison on long-form video understanding benchmarks with Qwen2.5-VL-7B at 64 and 128 input frames (%). At the 50% retain ratio, ST-SimDiff matches the full-token upper bound on VideoMME and comes close on LongVideoBench, confirming that its effectiveness extends to this additional model architecture.

To further establish the generalization of our framework, we conduct additional experiments on the recently released Qwen2.5-VL model, evaluating on VideoMME and LongVideoBench to confirm that the method’s efficacy is not tied to a particular model architecture but transfers broadly.

The results in Table 4 attest to the strong performance of ST-SimDiff on Qwen2.5-VL. At a 30% token retention ratio, our method scores 62.4 on VideoMME and 56.8 on LongVideoBench; raising the budget to a 50% retention ratio lifts these to 62.9 and 58.4, respectively. Notably, at the 50% ratio the VideoMME score matches the upper bound obtained with the full token set, and the LongVideoBench score is closely comparable to its full-token counterpart. These findings affirm the strong generalization of our framework across model architectures.

| Method                        | VideoMME    | LongVideoBench | EgoSchema   |
|-------------------------------|-------------|----------------|-------------|
| Token Retain Ratio $r = 30\%$ |             |                |             |
| Baseline                      | 59.3        | 53.8           | 50.2        |
| + Sim                         | Spatial     | 59.5           | 54.5        |
|                               | Temporal    | 59.6           | 54.3        |
|                               | Spa. + Tem. | 59.9           | 54.6        |
| <b>++ Diff (Proposed)</b>     | <b>60.2</b> | <b>55.2</b>    | <b>51.7</b> |
| Token Retain Ratio $r = 50\%$ |             |                |             |
| Baseline                      | 60.9        | 55.3           | 51.5        |
| + Sim                         | Spatial     | 61.1           | 55.7        |
|                               | Temporal    | 61.3           | 55.6        |
|                               | Spa. + Tem. | 61.4           | 56.0        |
| <b>++ Diff (Proposed)</b>     | <b>61.7</b> | <b>56.5</b>    | <b>52.5</b> |

Table 5: Ablation of the framework’s components on the NVILA-Video-8B model (%). The trends mirror those observed on LLaVA-Video-7B: the similarity module (+ Sim), its spatio-temporal joint variant (Spa. + Tem.), and the difference module (++ Diff) each contribute consistent gains, with the complete framework best at both retain ratios.

## C ABLATION STUDY RESULTS ON NVILA

To establish that each component generalizes across base models, we replicate our detailed ablation on the NVILA-Video-8B model (Liu et al., 2024c). The setup follows that of the main-paper ablation: starting from a Baseline that performs only fundamental importance-based pruning, we progressively add the similarity selection module (+ Sim) and the difference selection module (++ Diff).

As Table 5 shows, the results reaffirm the framework’s design and track closely with the trends observed on LLaVA-Video-7B. Adding the similarity module (+ Sim) on top of the Baseline yields a steady improvement at both the 30% and 50% token retain ratios. Within this module we compare three variants—spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.)—and the joint variant proves most effective: at  $r = 50\%$ , for example, it reaches 61.4% on VideoMME, surpassing both the spatial-only and temporal-only strategies. This again underscores the importance of modeling spatio-temporal associations jointly in order to identify redundant information effectively. Adding the difference selection module (++ Diff) on top of the optimal spatio-temporal joint similarity module completes the ST-SimDiff framework and, as the table makes clear, produces the largest gains. This outcome demonstrates the effectiveness and cross-model generalization of the framework’s two core modules—similarity and difference selection—and again confirms our central idea of handling redundancy and change simultaneously.

## D ABLATION RESULTS OF $\tau_{sim}$ AND $\tau_{diff}$ ON LLaVA-VIDEO

The similarity threshold  $\tau_{sim}$  and the difference threshold  $\tau_{diff}$  are the two key hyperparameters of our framework, governing the selection of representative and transitional tokens respectively. To characterize their effect, we ablate each in turn, with results in Figure 6. Both parameters induce the same qualitative trend—performance rises and then falls around an optimum—while exhibiting strong robustness across a broad surrounding range. For  $\tau_{sim}$ , too low a value yields imprecise community detection, whereas too high a value fragments the integrity of the semantic clusters; for  $\tau_{diff}$ , too low a value admits noise through over-sensitivity, whereas too high a value causes key events to be missed. The model attains its optimum at  $\tau_{sim} = 0.8$  and  $\tau_{diff} = 0.2$ , and we accordingly adopt this configuration throughout the remaining experiments.

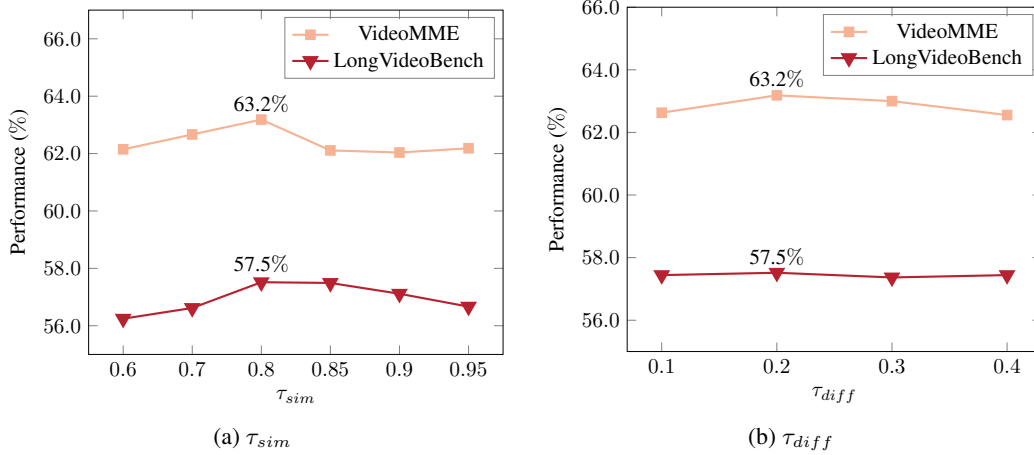


Figure 5: Ablation over the values of  $\tau_{diff}$  and  $\tau_{sim}$  on VideoMME and LongVideoBench. Performance rises and then falls around the optimum for both thresholds, and remains stable across a broad surrounding range.

## E ABLATION RESULTS OF $\tau_{sim}$ AND $\tau_{diff}$ ON NVILA-VIDEO

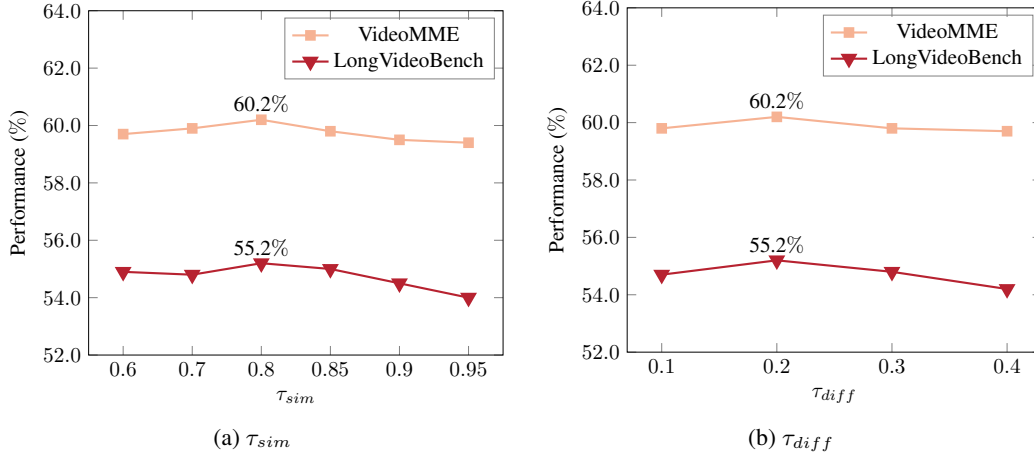


Figure 6: Ablation over the values of  $\tau_{diff}$  and  $\tau_{sim}$  on NVILA-Video-8B. As on LLaVA-Video, both thresholds peak at the same operating point and vary only marginally across a wide range, evidencing model-agnostic robustness.

To characterize the robustness of our key hyperparameters on a second architecture, we repeat the threshold ablation on the NVILA-Video-8B model. As Fig. 6 shows, we evaluate the effect of the similarity threshold  $\tau_{sim}$  and the difference threshold  $\tau_{diff}$  on performance across VideoMME and LongVideoBench.

**Impact of Similarity Threshold  $\tau_{sim}$ .** Sweeping  $\tau_{sim}$  over  $[0.6, 0.95]$ , we find performance to be largely insensitive to the threshold. On VideoMME it peaks at 60.2% at  $\tau_{sim} = 0.8$ , and on LongVideoBench it is likewise optimal at  $\tau_{sim} = 0.8$ , reaching 55.2%. Although performance dips slightly when the threshold is set too high or too low, the overall fluctuation is minimal, confirming that our method remains stable across a wide range of similarity thresholds.

**Impact of Difference Threshold  $\tau_{diff}$ .** Sweeping  $\tau_{diff}$  over  $[0.1, 0.4]$ , we again observe strong robustness. On both benchmarks the optimum is attained near  $\tau_{diff} = 0.2$ —60.2% on VideoMME

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and 55.2% on LongVideoBench—indicating that a difference threshold around 0.2 most effectively captures key events, maximizing performance under token compression.

In sum, these studies confirm the robustness of our method to its key hyperparameters and validate the choice of  $\tau_{\text{sim}} = 0.8$  and  $\tau_{\text{diff}} = 0.2$  in the main experiments, values that consistently yield optimal performance across benchmarks and base models.